

Chapter 1

Mathematical Problem Solving

1.1 Introduction

Mathematical problem solving has been established as the central theme of the Singapore primary and secondary mathematics curriculum since the 1990s. The primary aim of the curriculum is to develop students' ability to solve mathematics problems.

Curriculum documents state that problem solving heuristics are to be taught in our primary and secondary school mathematics classes. Singapore teachers have been provided with preservice preparation or professional development in teaching problem solving, drawing on overseas resources as well as those developed locally. However, these resources tend to emphasise the learning of heuristics but do not focus on mathematics content at a deep level or the kind of mathematical thinking used by mathematicians such as conjecturing and proving.

Towards upper secondary and junior college levels, on the other hand, students tend to concentrate on national exam-type mathematics problems, and in our experience, the heuristics learned at lower levels tend to be ignored by both students and teachers instead of being applied in their mathematical engagements.

To return problem solving to its focal role in the curriculum, we need to build teachers' capacity for problem solving and its teaching, and to develop materials and teaching resources for the explicit teaching of problem solving in the environment of deep mathematics content at the higher school levels. This booklet is a contribution towards this goal. It

presents a complete basic problem solving module which can be implemented in a high school curriculum.

The module materials consist of the following:

- Ten one-hour lesson plans
- Templates for a practical worksheet and an assessment rubric
- A collection of problems with
 - accompanying complete solution(s),
 - heuristics to be highlighted,
 - Pólya's stages to be highlighted,
 - suitable scaffolding that a teacher can give to the student,
 - suggested adaptations, extensions and generalizations to the problem,
 - anticipated student responses, and
 - notes on assessment of and for problem solving.

As a necessary preamble to the module materials, we shall first introduce Pólya's model of problem solving via a worked example in Section 1.2. In Section 1.3, we then gird this model with the framework of Schoenfeld which helps to bring into focus the different aspects that contribute to success in problem solving. Finally, in Section 1.4, we explain our approach to teaching problem solving via a 'practical' paradigm.

1.2 Pólya's Model of Problem Solving

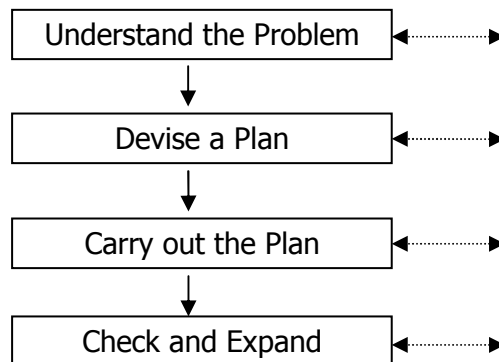
We would suggest that any attempt at mathematical problem solving requires a model to which the problem solver can refer, especially when he or she is unable to progress satisfactorily. Good problem solvers have intuitively built up their own problem solving models.

However, for two reasons, a problem solving model that is made explicit would be immensely helpful. Firstly, the typical problem solver who has no model may find that a model helps in regulating his problem solving attempt. Secondly, even a good problem solver will

find the structured approach of a model useful and an earlier introduction to the process will help him progress faster in his mathematical development. Consider what Alan Schoenfeld, a mathematician and mathematics educator has to say in the preface to his book *Mathematical Problem Solving*:

In the fall of 1974 I ran across George Pólya's little volume, *How to Solve It*. I was a practising mathematician, a few years out of graduate school and happily producing theorems in topology and measure theory. Pólya wrote about problem solving, more specifically about the strategies used by mathematicians to solve problems ... My first reaction to the book was sheer pleasure. If, after all, I had discovered for myself the problem-solving strategies described by an eminent mathematician, then I must be an honest-to-goodness mathematician myself! After a while, however, the pleasure gave way to annoyance. These kinds of strategies had not been mentioned at any time during my academic career. Why wasn't I given the book when I was a freshman, to save me the trouble of discovering the strategies on my own?

We shall now describe the essential features of the problem solving model proposed by George Pólya, an eminent Hungarian mathematician. The process can be summarized in the diagram below.



The model resembles a flowchart with four components, *Understand the Problem*, *Devise a Plan*, *Carry out the Plan*, and *Check and Expand*, which are hierarchical but which allow back-flow. We shall explain the components of the model and how the model works by applying it to a problem as an example.

The Lockers Problem

The new school has exactly 343 lockers numbered 1 to 343, and exactly 343 students. On the first day of school, the students meet outside the building and agree on the following plan. The first student will enter the school and open all the lockers. The second student will then enter the school and close every locker with an even number. The third student will then ‘reverse’ every third locker; i.e. if the locker is closed, he will open it, and if the locker is open, he will close it. The fourth student will reverse every fourth locker, and so on until all 343 students in turn have entered the building and reversed the relevant lockers. Which lockers will finally remain open?

Understand the Problem

First, one must realize that such a scenario could hardly happen in real life, simply because it is not possible for 343 kids to wait patiently for their turn to do such a tedious job! Thus, the problem is one of ‘fantasy’ or ‘cognitive playfulness’. The key parts of the problem to take time to understand would be:

- that to ‘reverse’ a locker means to open it when it is closed and to close it when it is open;
- which lockers would a particular student act on.

It is common to see students staring at a blank piece of paper when they are stuck at a problem. The problem is too difficult for him or sometimes even for him to understand and so he cannot start. It is important that something is done and the following describes how *heuristics* (meaning: serving to find out or discover, *Oxford English Dictionary*) can help start the problem-solving process.

Two useful *heuristics* to help understand the problem are to *act it out* and to *consider a simpler problem (smaller numbers)*. Consider the problem when there are only 10 students. We list quickly the 10 lockers as 1, 2, ..., 10 on a piece of paper. Now we act out what the 10 students will do, using the following *representation*: a backslash '/' for open and a slash '/' over the backslash to obtain a cross '×' for close. The following shows what will result after a few students have been acted out (the actions taken in each step are in bold):

- 1) 1 2 3 4 5 6 7 8 9 10
 / / / / / / / / / /
- 2) 1 2 3 4 5 6 7 8 9 10
 / × / × / × / × / ×
- 3) 1 2 3 4 5 6 7 8 9 10
 / × × × / ×/ / × × ×
- 4) 1 2 3 4 5 6 7 8 9 10
 / × × ×/ / ×/ / ×/ × ×

The problem (not the solution) looks clear now. If we want to, we could solve the simplified problem for 10 students. This may be our first plan.

Note: Schoenfeld expands Pólya's model in his book *Mathematical Problem Solving* and suggests the following to try to understand the problem.

Transcribe the problem in shorthand, drawing a diagram or making a rough sketch if you can. If someone took away the original problem statement, would you have enough to work with? Have you really identified all the essential information in the problem statement?

Devise a Plan

A plan is vital to the success of any endeavour. It has been said, "He who fails to plan, plans to fail."

It is common, when asking a student who is stuck at a problem what he is doing, to receive the reply “I don’t know.” Even though he may be scribbling something on the paper (which is usually better than nothing), there is no plan driving his effort. It is important for the problem solver to be able to articulate his plan so that he is guided by it.

Back to our problem, our first plan is: we will *consider a simpler problem (smaller numbers)* for 10 students, and *look for patterns*.

Carry out the Plan

With the plan decided, carrying it out requires the skills and knowledge that should be automatic in the problem solver. If there is a severe lack of resources, such as content knowledge, algebraic skills or mathematical techniques, the best plan will go awry. Thus, it is very important in teaching mathematical problem solving to know that a sufficiently strong foundation in mathematics knowledge is often necessary for success. In this case, the student must know the concept of multiples of numbers very well for efficiency.

So let us continue:

$$5) \quad \begin{array}{cccccccccc} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 \\ / & \times & \times & \times/ & \times & \times/ & / & \times/ & \times & \times/ \end{array}$$

$$6) \quad \begin{array}{cccccccccc} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 \\ / & \times & \times & \times/ & \times & \times \times & / & \times/ & \times & \times/ \end{array}$$

$$7) \quad \begin{array}{cccccccccc} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 \\ / & \times & \times & \times/ & \times & \times \times & \times & \times/ & \times & \times/ \end{array}$$

$$8) \quad \begin{array}{cccccccccc} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 \\ / & \times & \times & \times/ & \times & \times \times & \times & \times \times & \times & \times/ \end{array}$$

$$9) \quad \begin{array}{cccccccccc} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 \\ / & \times & \times & \times/ & \times & \times \times & \times & \times \times & \times/ & \times/ \end{array}$$

10)	1	2	3	4	5	6	7	8	9	10
	/	×	×	×/	×	×	×	×	×	×

Thus we have a solution for our simplified problem as ‘Lockers 1, 4 and 9 will remain open at the end’.

It will take different lengths of time for different persons to be able to find a pattern but the pattern once found has to be stated as a conjecture. It is very important to tell students that in mathematics, pattern is NOT proof.

Our conjecture after carrying out the initial plan is:

The lockers that will remain open will be those whose number is a square, i.e. when there are 343 students, they are 1, 4, 9, 16, 25, 36, 49, ..., 324.

Check

Returning to the original problem, we check our conjecture for the next square that we did not actually act out, i.e., 16. We note that we need only to look at the first 16 lockers (out of the original 343) since students numbered 17 and above will not touch the first 16 lockers, in general Student n will not touch lockers numbered less than n . We do so with the following result:

1	2	3	4	5	6	7	8	9	10	11	12	13
/	×	×	×/	×	×	×	×	×	×	×	×	×

14	15	16
×	×	×

The conjecture that the squares will remain open looks good so far!

Understand the Problem 2

We go back to Stage I to try to understand the problem deeper. Can we make progress by asking the question:

- What feature(s) of a square causes the corresponding locker to end up open? (Note that this is still a conjecture.)

The answer may not be obvious and so we ask a simpler question:

- By *acting it out* slowly, why does a particular locker end up open (or closed)?

By *acting it out* or by looking closely at the *representation* above, one will eventually realize that if a locker is ‘touched’ an odd number of times, it will end up open; if it is ‘touched’ an even number of times, it will end up closed.

Next, we use the heuristic *restate the problem in another way*.

- What type of number causes the corresponding locker to be touched an odd number of times?

Returning to the idea of features of a square, we ask:

- What feature(s) of a square causes the corresponding locker to be touched an odd number of times? (Note that this is still a conjecture.)

The series of questions involving the conjecture of squares, the lockers being touched an odd number of times, features of a square, distills to a more fundamental question which puts away the conjecture of squares for the moment.

- What feature(s) of a number causes the corresponding locker to be touched?

Again this may need different lengths of time for different persons. Eventually, we reach the understanding that a locker is touched only by students whose numbers are factors of the number of the locker. Thus, the number of times a locker is touched is exactly the number of the factors of its corresponding number.

Restating the problem in another way again:

- What type of number has an odd number of factors?

The series of questions using heuristics which twists, turns and focuses the original problem is crucial to better understand the problem before formulating a new plan of attack.

Remark: In some sense, the series of questions arise naturally and reflect the sequence of thinking that, hopefully, narrows down the original problem to one that can be solved.

Devise a Plan 2

Plan 2 will now be to count the number of factors of some numbers and try to understand why a square has an odd number of factors while the others have even numbers. We will *look for a pattern*.

Carry out the Plan 2

We will count the number of factors for 1 to 10 and then a few other random numbers when we spot a pattern. *Listing the factors systematically* will be helpful.

Number	Number of factors
$1 = 1$	1
$2 = 1 \times 2$	2
$3 = 1 \times 3$	2
$4 = 1 \times 4 = 2 \times 2$	3
$5 = 1 \times 5$	2
$6 = 1 \times 6 = 2 \times 3$	4

Number	Number of factors
$7 = 1 \times 7$	2
$8 = 1 \times 8 = 2 \times 4$	4
$9 = 1 \times 9 = 3 \times 3$	3
$10 = 1 \times 10 = 2 \times 5$	4
$16 = 1 \times 16 = 2 \times 8 = 4 \times 4$	5
$343 = 1 \times 343 = 7 \times 49$	4

It is now quite clear that a square has an odd number of factors while a non-square has an even number of factors – every factor of a number has a distinct ‘partner’, except for the root of the perfect square whose ‘partner’ is itself. At this point, we have essentially solved the problem as we now have the means—look for squares that have an odd number of factors—to find out if a locker will be open.

Devise a Plan 3

We will now write out a ‘clean’ solution focusing on showing that only the squares have an odd number of factors. We will do this by showing that the factors come in pairs.

Carry out the Plan 3

Let n be a positive integer. Let $a_1, a_2, \dots, a_k$ be all the k distinct factors of n with $a_1 < a_2 < \dots < a_k$. Observe the following:

- (i) n is the product of the i -th smallest factor a_i and the i -th largest factor a_{k+1-i} , i.e. $n = a_i a_{k+1-i}$, where $a_i \leq a_{k+1-i}$.
- (ii) If n is not a square, for $n = a_i a_{k+1-i}$, we have $a_i < \sqrt{n}$ and $a_{k+1-i} > \sqrt{n}$. This implies that each factor has a ‘partner’ factor distinct from itself such that the product of the two factors is n . Hence the number of factors must be even.
- (iii) If n is a square, then $\sqrt{n}$ is an integer. As in (ii), each factor less than $\sqrt{n}$ has a ‘partner’ factor distinct from itself such that the product of the two factors is n . Observe that $n = \sqrt{n} \times \sqrt{n}$ and so this factor $\sqrt{n}$ has no distinct ‘partner’. Hence the number of factors must be odd.

Now each locker will be ‘touched’ by a student whose number is a factor of the number of the locker. A square will be touched by an odd number of students because it has an odd number of factors as shown above. In the sequence of ‘open’ followed by ‘closed’, an odd number of actions will end in the locker eventually open. A non-square will, by the result above, be closed at the end.

Hence, the lockers that will remain open will be those whose number is a square, i.e. when there are 343 students, they are 1, 4, 9, 16, 25, 36, 49, ..., 324.

Check and Expand

We can check the solution for the locker 25. *Act out* the visitors to Locker 25, and as before make a backslash ‘/’ for open and a slash ‘\’

over the backslash to obtain a cross ‘ $\times$ ’ for close: 25 has factors 1, 5, 25 and so we have $\times/$, which means ‘open’.

We are done for this problem but not done with the problem-solving process. It is a key feature of the model that the solver should try to ‘expand’ the problem even though it is solved. By expanding, we mean one of the following:

- finding other solutions which are ‘better’ in the sense of elegance, succinctness, or with a wider applicability
- posing new problems
 - *adapting* by changing certain features of it (eg. change some numbers, change some conditions, consider the converse)
 - *extending* to problems which are more ‘difficult’ or which have greater scope
 - *generalizing* to problems which would include the given problem as a special example

We sketch an alternative solution that uses prime factorisation to prove that a natural number is a square if and only if it has an odd number of factors as follows.

Clearly, 1 has an odd number, 1, of factors.

Let $n > 1$ be a natural number and $n = p_1^{k_1} p_2^{k_2} \cdots p_s^{k_s}$ be its prime factorization. Let m be a factor of n . Then $m = p_1^{l_1} p_2^{l_2} \cdots p_s^{l_s}$ where $0 \leq l_i \leq k_i$ for all $i = 1, 2, \dots, s$.

By the Multiplication Principle, the number of possible m ’s, i.e., factors of n is $(k_1+1)(k_2+1)\dots(k_s+1)$. This product is odd if and only if each of its factors (k_i+1) is odd, which in turn happens if and only if each k_i is even. Thus the number of factors of n is odd if and only if n is a square.

We pose the following problems with sketches of their corresponding proofs.

Adaptation 1: The i -th student reverses every i -th locker except Locker i .

Sketch of solution: Now, every locker is touched as many times as its number of factors less one. Thus, a non-square is touched an odd number of times and so remains open at the end.

Adaptation 2: The i -th student reverses every locker whose number is a factor of i .

Sketch of solution: The multiple-factor relationship in the original problem is now reversed. Thus, we are interested in the number of multiples of a number m . The locker with number m will remain open at the end if and only if $\left\lfloor \frac{343}{m} \right\rfloor$ is odd.

Extension: There are m stages to finally open the locker (for example: put the key in the lock, turn the key, pull down the handle, etc.). The i -th student goes to every i -th locker and acts out the next stage, and if the locker is fully open, closes and locks it. Which lockers remain open in the end?

Sketch of solution: Again, every locker is touched as many times as its number of factors but now, it remains open if and only if its number of factors is $m \bmod (m+1)$.

Generalisation 1: There are n lockers. How many lockers remain open?

Sketch of solution: The lockers that remain open are the squares which do not exceed n . Number of lockers that remain open = $\left\lfloor \sqrt{n} \right\rfloor$.

1.3 Focusing with Schoenfeld's Framework

We continue with Schoenfeld's comments in his preface to *Mathematical Problem Solving*.

The next day I spoke to the colleague who trained our department's team for the Putnam exam, a prestigious

nationwide mathematics competition. Did he use Pólya's book? "No," he said. "It's worthless." His teams did quite well, so there must have been some truth in what he said.

Schoenfeld grappled with the apparent worth of Pólya's model and the real-world failure of its application in the classroom. His research led him to realize that there was more than just a direct application of the model and that in fact, other factors are crucial in successful problem solving. His research culminates in the construction of a framework for the analysis of complex problem-solving behaviour. The framework, in his words, consists of four aspects:

1. Cognitive resources – the body of facts and procedures at one's disposal
2. Heuristics – 'rules of thumb' for making progress in difficult situations
3. Control – having to do with the efficiency with which individuals utilise the knowledge at their disposal
4. Belief systems – one's perspectives regarding the nature of a discipline and how one goes about working on it

With regard to our attempts to teach problem solving, Schoenfeld's framework makes us realize that Pólya's model or the teaching of heuristics is not all there is. The problem solver needs also to address the availability of resources and exercise cognitive and affective control over the problem solving process. The framework allows teachers to have a structure through which they can focus when difficulties arise in solving a problem. We use our example to illustrate how Schoenfeld's framework can be used.

Cognitive resources

With respect to the original locker problem, failure will result if there is a lack of knowledge of any of these: 'reverse' or 'every fourth' in *Understanding the Problem*; 'multiples' and 'factors' in *Devise a Plan* and *Carry out the Plan*; 'prime factorisation' for the alternative proof in *Check and Expand*.

Our study of problem solving (Quek, Choy, Dong, Toh & Ho, 2007) suggests that we regard *resources* as necessary but not sufficient for success in problem solving. Further, we recommend that teachers should not neglect the provision of *resources* during the planning of curriculum and the teaching of mathematics. It is not viable—at least in the school situation they were in—to devote a substantial amount of curriculum time meant for important content and concepts of mathematics and their interrelationships to just teach processes, i.e. Control and Heuristics, or ‘motivate’ students, i.e. change Beliefs.

Heuristics

Many problem solving programmes emphasise the use of heuristics. Schoenfeld is no exception. The use of heuristics is the most visible facet of successful problem solving. For the observer, it would seem that the application of a suitable heuristic unlocks the puzzle. Schoenfeld initially saw the centrality of heuristics in Pólya’s model. Later, he realized that the other three less visible aspects of resources, control and beliefs must be taken together to obtain a fuller picture of successful and unsuccessful problem solving.

In our example, heuristics are used to explore the problem (*act it out* and *consider a simpler problem (smaller numbers)*), as the basis of a plan (*consider a simpler problem (smaller numbers)* and *look for patterns*), and while carrying out the plan (*use a suitable representation (table)*).

A list of heuristics that we will use in this book is:

- restate the problem in another way
- think of a related problem
- work backwards
- aim for sub-goals
- divide into cases
- use suitable numbers (instead of algebra)
- consider a simpler problem
 - smaller numbers
 - special case - tighten conditions
 - fewer variables

- consider a more general case - loosen conditions
- act it out
- guess-and-check
- make a systematic list
- make a table
- look for patterns
- use equations/algebra
- draw a diagram
- use a suitable representation
- use suitable notation

Control

Control, also called metacognition, is a major determinant of problem-solving success or failure (Schoenfeld, 1985). In our opinion, much of the difficulty in teaching problem solving or any other aspect of mathematics in our schools arises from underemphasizing the role of metacognition. The teaching of heuristics *per se* often takes place in place of teaching problem solving with all its different aspects. To be fair, the teaching of control during problem solving is very difficult.

In our example, control is needed in the second round of understanding the problem where the series of questions asked finally focused on the key aspect of factors. The good problem solver will ‘sense’ that he cannot proceed with a solution until he is sure of why the lockers with square numbers remain open at the end. He will ‘force’ himself to apply some heuristics to make sure that the problem is well understood before rushing on to work on it.

Control is needed when devising a plan. Again, the good problem solver is patient and willing to go through more than one plan, choosing a modest goal first before launching into a full scale attack on the problem. Many students with poor control end up spending more time looking for a quick solution – to them, ‘quick’ means a ‘magic bullet’ of a formula that will give the answer.

Control is needed when carrying out the plan, to decide on a suitable representation (the slash and cross) in marking out the actions of opening and closing in Plan 1, and to decide how many operations to continue doing in a search for a pattern in Plan 2. Control is constantly needed to monitor if we are still on the right track when algebraic manipulations and numerical calculations are involved. In fact, control is necessary to regulate the starts and stops, turns and returns of the problem solving journey.

Even good students do not usually want to extend a problem, being satisfied with the successful solving of the original problem. Such students fall just short of developing a mature mathematical way of thinking. Control is needed when checking and extending. In some sense, this control is for the larger arena of *all* problems in that the insights gained from following through with Pólya's fourth and last stage will 'save time' for future problems. In our example, the adaptations offer reinforcement for the method used and the alternative solution offered a number-theoretical perspective.

Control also means knowing when to store a problem for another time, or to give it up permanently or temporary, to regurgitate at a suitable time. For some students, we may encourage them to take even days to work out the relevant questions of Understanding the Problem 2.

Beliefs

If a person believes that real life does not need mathematics or at best, that mathematics should be left to a few mathematicians, then it will not be surprising that, that very person will not find it of any value to solve 'difficult' mathematics problems. If a person believes that there is always a fixed method to solve mathematics problems and anything else beyond that belongs to the realm of the nerds (again), then such a person will give up whenever he encounters a mathematics problem that cannot be solved within a short time.

Beliefs are strong and ingrained in a person and beliefs about mathematics strongly affects how one approaches mathematics and mathematics problems.

In our example, a person who believes that mathematics problems are to be solved ‘in a short period of time or not at all’ will give up quickly when acting out the opening and closing of the lockers. Again, a person who believes that Mathematics is all about formula may likely give up after realizing that there is no ‘magic bullet’ formula for this problem.

A person who knows that listing is a worthwhile heuristic to explore a problem or to solve part of it perseveres even though the listing may seem slow, tedious and to reward little. A different person may believe that ‘listing is like a train - it starts slowly but picks up speed’ and thus, perseveres knowing that the initial cases though slow do not translate into proportional time when extended to the general case. Thus, after spending quite some time slowly recording the first few students acting on the lockers, the rest would be a breeze.

In summary, successful problem solving is not a mere application of some model. It is the complex interaction of at least four components (Resources, Heuristics, Control and Beliefs). On the flip side, lack of success in problem solving can be attributed to a haphazard approach lacking one or more of the positive aspects of the said components.

1.4 The Practical Paradigm

1.4.1 Evolution of Problem Solving instruction

As we taught some problem solving classes, we noticed that students were resistant to following the stages of Pólya’s model. Although we coaxed and encouraged, generally, students did not take to the model. Even the high ability mathematics students who could solve the given problems refused to make the extra effort to finally Check and Expand the problem.

For them, although there was apparent progress in their ability to use heuristics, the metacognitive part of the problem solving process still left much to be desired. Schoenfeld (1985) asked out loud specific control questions at regular intervals as a way to make students pause and be more aware of their thinking.

Sometimes, it is good to steal a sideways glance at what others are doing. Tay (2001) discussed the use of Cloze and comprehension passages to teach and assess the skills of reading and writing mathematics. The paradigm for this came from the teaching of language. In an attempt to 'make' the students follow the Pólya model, especially when they were clearly struggling with the problem, we decided to construct a worksheet (described below) like that used in science practical lessons and told the students to treat the problem solving class as a mathematics 'practical' lesson. In this way, we hoped to achieve a paradigm shift in the way students looked at these 'difficult, unrelated' problems which had to be done in this 'special' class. The science practical lesson is very much accepted by students as part of science education and many have an understanding that it is to teach them how to 'do' science. Woolnough and Allsop (1985) stated clearly what is to be achieved in science education.

As we look at the nature of science we see two quite distinct strands. The knowledge, the important content and concepts of science and their interrelationships, and also the processes which a scientist uses in his working life. In teaching science we should be concerned both with introducing students to the important body of scientific knowledge, that they might understand and enjoy it, and also with familiarizing students with the way a problem-solving scientist works, that they too might develop such habits and use them in their own lives.

(Woolnough and Allsop, 1985, p.32)

It is instructive to see that we could just replace 'science' with 'mathematics' and the preceding passage reads just as truly. Practical work to achieve the learning of the scientific processes has a long history

of at least a hundred years and can be traced to Henry Edward Armstrong (Woolnough and Allsop, 1985; Armstrong, 1891). Despite much debate of how exactly it is to be carried out (Woolnough and Allsop, 1985), practical work is accepted as a mainstay in science education. It is certainly conceivable that similar specialised lessons and materials for mathematics may be necessary to teach the mathematical processes, including and via problem solving. To this, we propose the idea of Mathematics Practical.

Thus, in a 55-minute Mathematics Practical lesson (we assume that each lesson has a duration of 55 minutes), the class focuses on one problem and students have to work this out on a Mathematics Practical worksheet.

The worksheet consists of 4 pages (see Appendix 1), each page corresponding to one Pólya stage. Ideally, the student will follow the model and go through all 4 stages, with suitable loopbacks (work on blank piece of paper and attach at the relevant place).

However, so as not to straitjacket an unwilling problem solver, the student may jump straight to Stage 3: Carry out the Plan and is given up to 15 minutes to solve the problem there. A student who completes the problem in time needs only do the last Pólya stage. If not, it is explained to him that the Pólya model would allow him to do better than what he has achieved and so he is required to go through all four Pólya stages.

These alternatives are intended to differentiate between the needs of the different problem solvers, where the unsuccessful student is required to systematically and metacognitively go through the Pólya model and the successful student is allowed to leapfrog to the fourth stage which is important for consolidation of the method and gaining a fuller insight to the problem.

Certainly, a particular student may have to go through all 4 stages for one problem and not for another problem where the plan to him is obvious and he needs only to Check and Expand. The choice allowed here is to show that explicit use of Pólya's model is not necessary all the time but

would be very useful when one is stuck and when one is learning to problem-solve.

Appendix 1 is a template for the Practical Worksheet and Appendix 2 is a scan of work done on the Lockers Problem using the Practical Worksheet by a graduate of the problem solving module.

1.4.2 Choosing problems for the Mathematics Practical

The successful implementation of any problem solving curriculum hinges on the choice of *appropriate* problems. However, *appropriate* problems may have different connotations for different people.

The nurturing of problem solving skills requires students to solve meaningful problems. Lester (1983) claimed that posing the cleverest problems is not productive if students are not interested or willing to attempt to solve them. The implication is that mathematical problems have to be chosen judiciously. What should be the criteria for choosing good problems? Problems selected for a course must satisfy five main criteria (Schoenfeld, 1994, cited in Arcavi, Kessel, Meira, & Smith, 1998, pp. 11-12):

1. Without being trivial, problems should be accessible to a wide range of students on the basis of their prior knowledge, and should not require a lot of machinery and/or vocabulary.
2. Problems must be solvable, or at least approachable, in more than one way. Alternative solution paths can illustrate the richness of the mathematics, and may reveal connections among different areas of mathematics.
3. Problems should illustrate important mathematical ideas, either in terms of the content or the solution strategies.
4. Problems should be constructible or solvable without tricks.
5. Problems should serve as first steps towards mathematical explorations, they should be extendable and generalizable; namely, when solved, they can serve as springboards for further explorations and problem posing.

Conscious of the fact that the choice of problems was critical in our problem solving project, we were guided by the following principles: (1) the problems were interesting enough for most if not all of the students to attempt the problems; (2) the students had enough “resources” to solve the problem; (3) the content domain was important but subordinate to processes involved in solving it; and (4) the problems were *extendable* and *generalizable*.

Guided by these criteria, we have chosen problems for the Mathematics Practical lessons (Chapter 4). Some of these choices of the problems were based on data collected from previous students solving these problems. The authors are in the process of developing three additional sets of corresponding mathematics problems in Chapter 4. These problems will be available in the near future.

1.4.3 Scaffolding in the Mathematics Practical

What does the teacher do when students are working on a problem using the Practical Worksheet? As in the science practical where the teacher or technician provides help in setting up apparatus and in understanding the experimental setup and analysis with the view to learn the processes of science, the mathematics teacher should be clear that his overt help is towards improving the problem solving processes of the student seeking help.

To this end, we propose three levels of help that a teacher can give to a student. The levels are hierarchical and one level should be given only after an earlier level has failed. In the crucial level which we call Level 0, we emphasise the student learning and reinforcing the Pólya model. We may ask the student if he knows what Pólya stage he is in, and what would one normally do in such a stage. We help by asking the two Control questions. In Level 1, we suggest *specific* heuristics to get the work moving. Level 2 is to be avoided as much as possible and is included only for the important aspect of ensuring that the self-esteem of the student is not seriously damaged by his perceived failure and helplessness on the problem. Here, we give problem specific hints, which essentially is a throwback to the ‘usual’ help afforded by mathematics teachers. The

objective of the practical paradigm is for students to internalize Level 0, and ask for Level 1, and to a much lesser extent Level 2, hints only when pressed for time.

Level	Feature	Examples based on the Lockers Problem
0	Emphasis on Pólya stages and control	What Pólya stage are you in now? Do you understand the problem? What exactly are you doing? Why are you doing that?
1	Specific heuristics	Why don't you try with fewer lockers (<i>use smaller numbers</i>)? <i>Try looking for a pattern.</i>
2	Problem specific hints	Think in terms of the locker rather than the student – what numbers get to touch the locker?

The set of problems in Chapter 4 are placed within the template of the Practical Worksheet and include suggestions for scaffolding the students. Do take note that Level 0 prompts are not specifically stated. The teacher must always remember to first ask Level 0 questions; he may also direct the student to respond to the instructions and questions in the Practical Worksheet template as they are essentially Level 0 questions.

1.4.4 Assessment of the Practical Worksheet

Meeting the challenge of teaching mathematical problem solving to students, calls for a curriculum that emphasises the process (while not neglecting the product) of problem solving and an assessment strategy to match it, to drive teaching and learning. Effective assessment practice “begins with and enacts a vision of the kinds of learning we most value for students and strive to help them achieve” (Walvoord & Anderson, 1998).

Correspondingly it is common knowledge that most students will not

study for curricular components which are not assessed. We have seen how the Practical Worksheet fits into the paradigm of a Mathematics Practical. We now present the scoring rubric as an assessment instrument for the Practical Worksheet.

The scoring rubric focuses on the problem-solving processes highlighted in the Practical Worksheet. There are four main components to the rubric, each of which would draw the students' (and teachers') attention to the crucial aspects of as authentic as possible an attempt to solve a mathematical problem:

- Applying Polya's 4-phase approach to solving mathematics problems
- Making use of heuristics
- Exhibiting 'control' during problem solving
- Checking and expanding the problem solved

In establishing the criteria for each of these facets of problem solving, we ask the question, "What must students do or show to suggest that they have used Polya's approach to solve the given mathematics problems, that they have made use of heuristics, that they have exhibited 'control' over the problem-solving process, and that they have checked the solution and extended the problem solved (learnt from it)?"

The rubric is presented below. The total score that a student is able to score for a problem is 20 marks.

- *Polya's Stages* [0-7] – this criterion looks for evidence of the use of cycles of Polya's stages (Understand Problem, etc).
- *Heuristics* [0-7] – this criterion looks for evidence of the application of heuristics to understand the problem, and devise/carry out plans.
- *Checking and Expanding* [0-6] – this criterion is further divided into three sub-criteria:
 - Evidence of checking the correctness of solutions [1]
 - Providing for alternative solutions [2]
 - Adapting, extending and generalizing the problem [3] – full marks for this are awarded for one who is able to

provide (a) two or more generalizations of the given problem with solutions or suggestions to solution, or (b) one significant extension with comments on its solvability.

The rubric is designed to encourage students to go through the Polya stages when they are faced with a problem, and to use heuristics to explore the problem and devise a plan. They would return to one of the first three phases (see the Practical Worksheet in Appendix 1) upon failure to realize a plan of solution. Students who ‘show’ control (Schoenfeld’s framework) over the problem solving process earn marks. For example, a student who did not manage to obtain a correct solution would be able to score up to five marks each for *Polya’s Stages* and for *Heuristics*, making a total of ten, if they are show evidence of cycling through the stages, use of heuristics and exercise of control.

The rubric allows the students to earn as many as 70% of the total 20 marks for a correct solution. However, this falls short of obtaining a distinction (75%) for the problem. The rest would come from the marks in *Checking and Extending*. Our intention is to push students to Check and Expand the problem, an area of instruction in problem solving that has been largely unsuccessful (Silver, Ghouseini, Gosen, Charalambous, & Strawhun, 2005).

Appendix 3 is a template of the Assessment Rubric.

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